

Data Analysis Write Up

Nov 01, 2000

The data analysis scheme looks as follows:

- Coda-2 runs at *adaqs3*.
- Raw data are written by CODA to */adaqs3/data2/moller_dataX_nnnn.dat*, where *X* is nothing or *X=_puls* or *X=_led* and *nnnn* is the run number.
- The raw data file should be converted to a NTUPLE file. The name and destination stay the same, but the extension is changed from *dat* to *nt*. There are 2 ways to do it.
 - **From PAW:** If one runs PAW (see later) on *adaql?* or *adaqs?* the file will be automatically converted when the appropriate run is attached by using *exec run run=nnnn*. Of course it is converted only once.
 - **Standalone.** Do:
 - Go to *moller* account on *adaqs?* or *adaql?*, any directory.
 - *rawrd.com -R nnnn*, where *nnnn* is the run number. One may type *rawdr.com -?* in order to get more information about this command.
- The NTUPLE file is analysed with PAW. While doing this one can attach up to 20 runs at a time and analyse them. When the next file (21-th) is attached one file (the first file attached) is disconnected and the new file is attached. The results of the data analysis are stored in local vectors of this PAW session. One can also calculate the average polarisation for a group of runs.
 - Login as "moller" to *adaql1* (or any other machine);
 - *bash* (it may work in *tcsh*, if the obsolete *PATH* declarations are fixed);
 - *cd paw/analysis* ;
 - *paw* (answer 3 for the workstation type);
 - *exec ini*
 - *exec run run=nnnn* (attaching the ntuple file for a run *nnnn*);
 - *exec lg_spectra icut=60 run=nnnn* (LG spectra), *cut=11* - coincidence only;
 - *exec asymr delay=0 angl=23. run=nnnn* (polarization calculation, there are many optional parameters);
 - *exec asym_av run1=n1 run2=n2* (averaging the polarization for the runs *n1-n2*);
 - *exec targ_angl runn1=nn1 runn2=n2 run1=n1 run2=n2 i=1/2/3* (calculating the target angle for the runs *n1-n2*, while the runs *nn1-nn2* were measured at 90°).
 - *call chngpol.f(p1,thet1,thet2)* recalculates the beam polarization for a different target angle *thet1*, *p1* is the polarization calculated at *thet2*.
 - *call prunpri.f(n1,n2)* (prints a table for the runs *n1-n2* is a technical parameter, namely an offset; it also writes a local file *runs.tab*)

Other useful tools in the PAW session:

- *exec sett_magp e0=3.25* prints the optimized GL values for the "Moeller" quads for a beam energy of 3.25GeV. Check the call parameters for other options.
- *mess \$call('spinpred.f(thet)')* prints the COS of the predicted spin angle for a Wien filter angle of *thet* (in degrees). Before, one should set a vector as: *ve/cr vprec(8) r 0.567 1 3*, where *0.567* is the linac's boost in GeV, *1* - is set for Hall A (2 - B ...), *3* - is the number of passes. One can also make a plot (like for spin dance): *fu/plot spinpred.f -180. 180.*
- If the ntuple creation starts before the data run is closed one has to:
 - PAW: *ld //* - check the LUN (say, LL) and the name (say XXXXX) of the file
 - PAW: *clo LL* - disconnect the file
 - PAW: *shell rm -i XXXXX* - delete the file
 - PAW: *ve/pri lrun* - find the element = run (say, the element NN)
 - PAW: *ve/in lrun(NN) 0*

- PAW: *ve/pri mrun* - find the element = LL (say, the element KK)
- PAW: *ve/in mrun(KK) 0*